Absolute Magnetic Permeability as a Characteristic of Minor Magnetic Properties

The technical description of the ferrite Fair-Rite 61 Material™ includes an interesting experimental characteristic—the dependence of the amplitude magnetic permeability μ_a on the amplitude of the magnetic flux density B. It is useful to examine this question if, in the future, we perform hysteresis-loop measurements ourselves.

Consider a cyclic symmetric minor loop nested within the major loop. Let the maximum values of magnetic field strength and magnetic flux density on this loop be denoted by H_{max} and B_{max} respectively. The relative amplitude magnetic permeability is defined by the following ratio (in SI units): $\mu_a = \frac{B_{max}}{\mu_0 H_{max}}$. For a family of symmetric minor loops with different excitation amplitudes, each pair of values $(H_{max,i}, B_{max,i})$ corresponds to a value $\mu_{a,i}$. The resulting set of points makes it possible to construct the dependence $\mu_a(B)$. For Fair-Rite 61 Material™ ferrite, the dependences $\mu_a(B)$ for two temperatures are shown in Fig. 1, where the minor loops were measured at a frequency of 10 kHz.

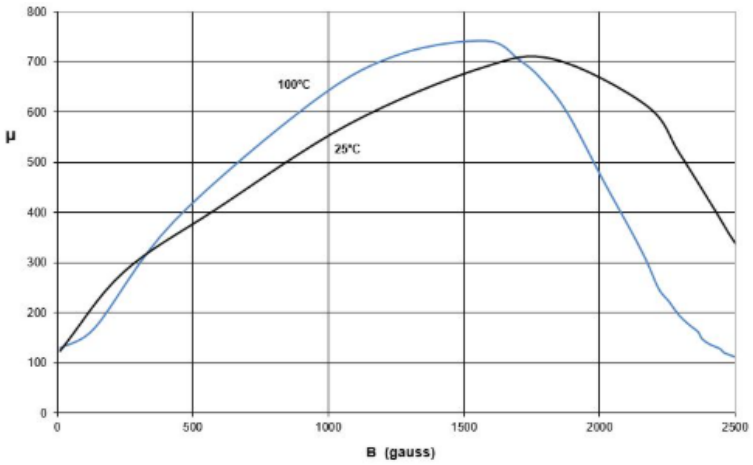


Fig. 1. Dependences of amplitude magnetic permeability on magnetic flux density for Fair-Rite 61 Material™ ferrite in cgs units.

Note that the experimental curve in Fig. 1 is not fully consistent with the major hysteresis loop given in the datasheet. A flux density of $B = 2500$ G, which is close to saturation, corresponds on the major loop to a field of $H \approx 13$ Oe; therefore, we should obtain $\mu_a = \frac{2500}{13} \approx 192.31$. However, the graph shows a value of ≈ 350 at 25 °C. The datasheet itself provides no explanation for these graphs. The measurement conditions for the minor loops used to calculate $\mu_a(B)$ should therefore be clarified.

The dependence of amplitude magnetic permeability on flux density shows the magnitude of magnetic field required to reach a specified flux density during symmetric cyclic magnetization reversal. Each point on this dependence corresponds to the apex of one symmetric hysteresis loop. The entire curve therefore contains information about the positions of the loop apices at different excitation amplitudes. This characteristic makes it possible to assess how correctly a model describes

magnetization reversal inside the major loop. Two models may reproduce the major loop equally well while predicting different minor-loop apices and, consequently, different amplitude permeability. At the same time, the positions of the apices do not determine the loop shape: the loop width, remanent flux density, and area may differ even when the apices coincide. Agreement with the experimental amplitude-permeability dependence is therefore an important, but not complete, validation of a minor-loop model.

Once the major loop has been specified, the Chan–Engelhardt and Tellinen–Madelung models do not use information about measured minor loops; instead, they generate them according to internal algorithmic rules that are physically motivated but remain model assumptions. In other words, the experimental data primarily determine the major loop of the material, whereas the minor loops in both models result from the selected construction mechanism. More complex models trained on measured minor loops also exist, but they are not considered in this report because this level of detail is not significant for our applications.

Nevertheless, we can construct a model dependence $\mu_a(B)$ and compare it with the experimental one, if available. To construct a model symmetric minor loop in the Tellinen–Madelung model, we start from a point (H_{start}, B_{start}) on the major branch and gradually reduce the amplitude of the periodic scanning field $H(t \geq 0) = (H_{max} + (H_{start} - H_{max})\exp(-t\,/\,(3T)))\cos(\omega t)$ to the selected limiting value $H(t \to \infty) = H_{max}$, where $T = 2\pi/\omega$, $\omega = 120\pi$, and $0 < H_{max} < H_{start}$. The scanning time is chosen sufficiently long for $H(t)$ to approach the limiting value H_{max}, after which the pair of limiting points (H_{max}, B_{max}) is used to calculate μ_a. Since an analytical formula for B_{max} cannot be specified, an evolutionary process is used. An example of a minor loop constructed by this evolutionary process is shown in Fig. 2. The digitized major loop of Fair-Rite 61 Material™ at 25 °C was used as the major loop.

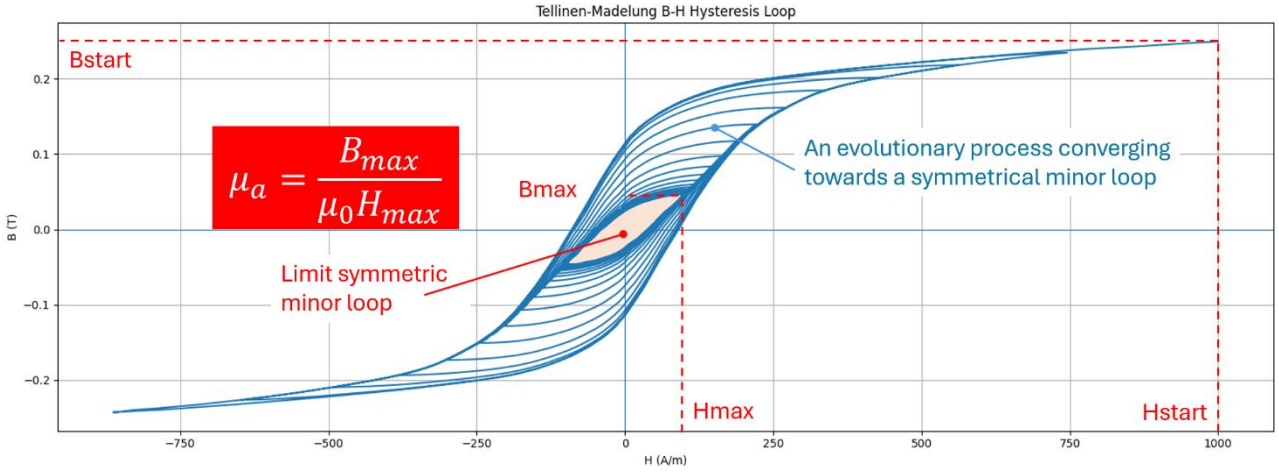


Fig. 2. Model symmetric minor loop obtained from the major loop of Fair-Rite 61 Material™ ferrite by periodic magnetization reversal with decreasing scanning-field strength.

The search for B_{max} is carried out over the last period of $H(t \geq 0)$. For the calculations, the major-loop data file “Major_loop.csv” must be provided with four columns (desc – descending, asc – ascending):

A	B	C	D
H_desc_A_per_m	B_desc_T	H_asc_A_per_m	B_asc_T
-1012.196906	-0.25	-1012.196906	-0.25
-954.0963435	-0.2475	-954.0963435	-0.2475
-903.6665154	-0.245	-903.6665154	-0.245

and the scanning-field parameters:

```
# ------------------------------------------------------------
# FIELD AMPLITUDE SCAN
# ------------------------------------------------------------

# Initial magnetic-field amplitude, A/m.
# Every calculation starts at H = +Hstart on the upper major branch.
Hstart=1000.0

# Final magnetic-field amplitude, A/m.
# Hb is scanned downward from Hstart to Hstop.
Hstop=10.0

# Number of Hb values in the interval [Hstop, Hstart],
# including both endpoints.
HbPoints=30
```

The resulting amplitude magnetic permeability dependence $\mu_a(B)$, calculated from the digitized major loop of Fair-Rite 61 Material™ at 25 °C, is shown in Fig. 3.

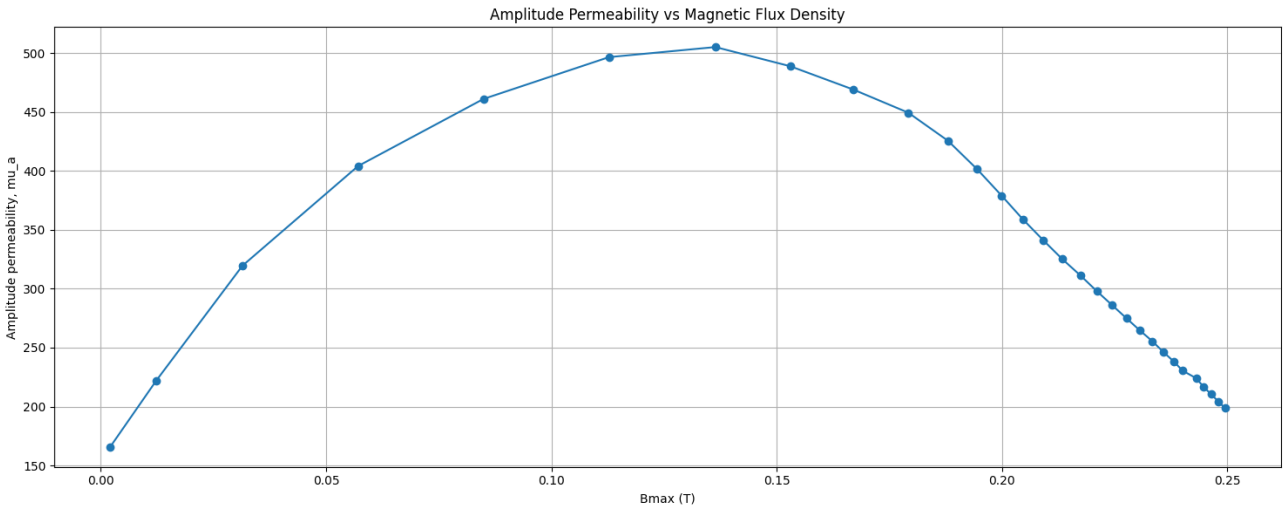


Fig. 3. Calculated dependence of amplitude magnetic permeability on flux density within the Tellinen–Madelung model, using the digitized major loop of Fair-Rite 61 Material™ at 25 °C.

In quantitative terms, it differs noticeably from the experimental dependence in Fig. 1: both the absolute level and individual features of the curve shape are different. At the same time, the overall character of the permeability variation with increasing magnetic flux density remains similar in both cases. However, a genuinely meaningful comparison of these dependences requires additional information about the method used for the experimental determination of the minor loops. Without knowing the excitation conditions, frequency, amplitude, and data-processing procedure, it is difficult to interpret the observed quantitative discrepancies unambiguously. In addition, as noted above, differences in the detailed shape of the minor loops appear to have only a weak effect on the current response of the inductor. Therefore, from the standpoint of circuit simulation, such discrepancies may be substantially less important than correctly reproducing the major loop, saturation, and the principal nonlinear properties of the magnetic material.

The comparisons presented above show that, in circuit-simulation problems, the detailed implementation of magnetic memory may have a substantially smaller effect on the electrical response than the correct description of the main nonlinear properties of the magnetic circuit: saturation, coercivity, remanent flux density, and physically plausible minor-loop behaviour. Therefore, for many electronics and electrical-engineering applications, there appears to be no need to pursue the most complex and formally elaborate hysteresis models if a simpler model adequately reproduces the observed currents and voltages. This is particularly important for calculations of the magnetic rapping actuator. It means that the result may be sufficiently robust with respect to a number of simplifying assumptions in the description of the magnetic circuit, including the use of approximate hysteresis models and a limited set of material parameters. In this case, the main attention should be focused on the factors that directly determine the force and transient response of the actuator: the geometry of the magnetic circuit, the magnitude and variation of the air gaps, saturation of individual core regions, winding resistance and parameters, and correct treatment of the external electrical circuit.

In the Chan–Engelhardt model, the branches of the major loop are defined by analytical template functions. Such functions can reproduce the shape of the loop of a real material with high accuracy. However, such agreement cannot be guaranteed in advance for every soft magnetic material. A more universal version of the model could be obtained by replacing the analytical template functions with experimental dependences $B(H)$, and then performing the subsequent minor-loop calculations using their interpolated representations. This is precisely the approach used in our Tellinen–Madelung calculations. Such a generalization would make the Chan–Engelhardt model more realistic for describing the magnetic properties of a particular material, especially the shape of its major loop. At the same time, the model would retain its important advantage: it would remain substantially less computationally demanding than Tellinen–Madelung because it does not require such a detailed description of magnetic memory, the nesting of previous states, or numerical integration.

The differences between Chan–Engelhardt and Tellinen–Madelung are manifested mainly in the way magnetic memory and the nesting of minor loops are described, but they do not necessarily prove significant for calculating inductor current. This practical insensitivity of the result to the details of the hysteresis model is of fundamental importance for engineering calculations. It gives reason to expect that computational models will remain predictive even when the material description is inevitably simplified. Otherwise, it would be difficult to speak of reproducible calculations and design solutions

for large electromagnetic devices such as transformers or plunger actuators if their characteristics depended strongly on subtle features of magnetic memory or on the particular procedure used to construct minor loops. For engineering modelling, it is important that the result be determined primarily by the main material characteristics, device geometry, and external-circuit parameters rather than by sensitive details of the selected hysteresis model. For this reason, the similarity of current responses obtained with different magnetic-hysteresis models should be regarded as a positive result. It shows that, in the operating regime considered, the inductor calculation has a certain robustness with respect to model assumptions and can therefore be used as a practical tool for further analysis of electromagnetic devices, including the magnetic rapping actuator.